

SMART IOT LIGHT CONTROL SYSTEM USING ESP32 AND BLYNK

CIRCUIT DIAGRAM:

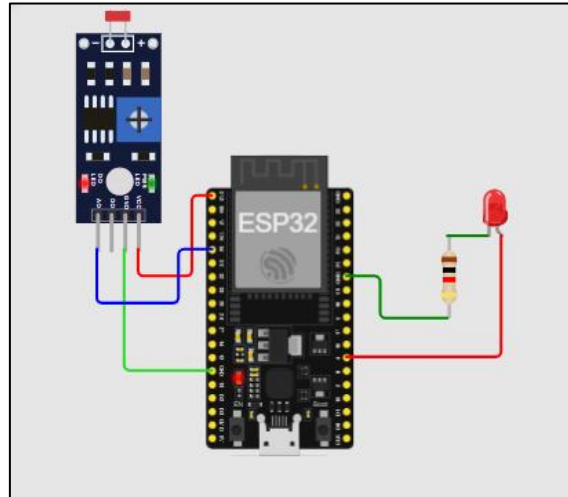


FIG.1

Components Used

- ESP32
- LDR Sensor Module (4-pin)
- LED
- 220 Ω Resistor
- Blynk Cloud Platform

Connections :

LDR Module Pin	ESP32 Pin
VCC	3.3V
GND	GND
AO	GPIO34
DO	Not Connected
LED Pin	ESP32 Pin
Anode (+)	GPIO4 through 220 Ω resistor
Cathode (-)	GND

PROJECT OVERVIEW

This project implements a Smart IoT Light Control System using ESP32, LDR sensor, and Blynk Cloud. The system can operate in both Manual Mode and Automatic Mode. Users can remotely monitor and control the light through the Blynk application.

HOW DECISION LOGIC WORKS :

The LDR sensor continuously measures the surrounding light intensity and sends the analog value to the ESP32.

When Auto Mode is enabled, the ESP32 compares the sensor reading with a predefined threshold value.

If the detected light intensity indicates a dark environment, the ESP32 automatically turns the LED ON.

If the detected light intensity indicates a bright environment, the ESP32 automatically turns the LED OFF.

Since the decision is made directly by the ESP32, the system continues to function even if there is a temporary delay in cloud communication.

DIFFERENCE BETWEEN MANUAL AND AUTO MODE

MANUAL MODE:

- The user controls the LED using the Blynk application.
- The LED turns ON or OFF according to the user's command.
- Internet connectivity is required for remote operation.
- Suitable when the user wants complete control over the lighting system.

AUTO MODE:

- The ESP32 controls the LED automatically using sensor data.
- No user intervention is required.
- Decisions are made locally based on ambient light conditions.
- Provides energy-efficient operation by turning the light ON only when needed.
- Faster response because the ESP32 does not wait for cloud commands.

SMART FEATURES IMPLEMENTED

○ SENSOR-BASED INTELLIGENT DECISION SYSTEM

The LDR sensor continuously monitors the surrounding light intensity and provides real-time data to the ESP32. The controller uses this information to determine whether the LED should be turned ON or OFF.

○ LOCAL AUTOMATION LOGIC

The ESP32 performs all decision-making locally. This eliminates dependency on cloud response time and ensures reliable operation even during network issues.

○ HYBRID CONTROL SYSTEM

The project supports both cloud-based and local control. Users can manually operate the LED through Blynk or enable Auto Mode to allow the ESP32 to control the LED automatically.

○ SMART IOT BEHAVIOR

The ESP32 sends sensor readings and LED status information to the Blynk dashboard. This allows users to monitor system performance remotely in real time.

SOURCE CODE

```
#define BLYNK_TEMPLATE_ID "TMPL30mtsdoEo"
#define BLYNK_TEMPLATE_NAME "Smart Light"
#define BLYNK_AUTH_TOKEN "_SEWsb776pRZ77qQibqqfbPGchX4IyTG"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
char ssid[] = "Wokwi-GUEST";
char pass[] = "";
#define LED_PIN 4
#define LDR_PIN 34
bool autoMode = 0, manualState = 0;
BlynkTimer timer;
BLYNK_WRITE(V0) {
    manualState = param.asInt();
    if (!autoMode) digitalWrite(LED_PIN, manualState);
}
BLYNK_WRITE(V1) {
    autoMode = param.asInt();
    if (!autoMode) digitalWrite(LED_PIN, manualState);
}
void sendData() {
    int light = analogRead(LDR_PIN);
    Blynk.virtualWrite(V2, light);
    if (autoMode)
        digitalWrite(LED_PIN, light > 2000);
    Blynk.virtualWrite(V3, digitalRead(LED_PIN));
}
void setup() {
    Serial.begin(115200);
    pinMode(LED_PIN, OUTPUT);
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
    timer.setInterval(1000, sendData);
}
void loop() {
    Blynk.run();
    timer.run();
}
```

OUTCOME

The Smart IoT Light Control System was successfully developed using ESP32, an LDR sensor, and the Blynk platform. The system operates in both manual and automatic modes, allowing users to remotely control the light or enable intelligent sensor-based automation. The ESP32 makes real-time decisions based on ambient light conditions, ensuring efficient operation. The project demonstrates IoT connectivity, local automation, cloud monitoring, and hybrid control, making it a practical and energy-efficient smart lighting solution.

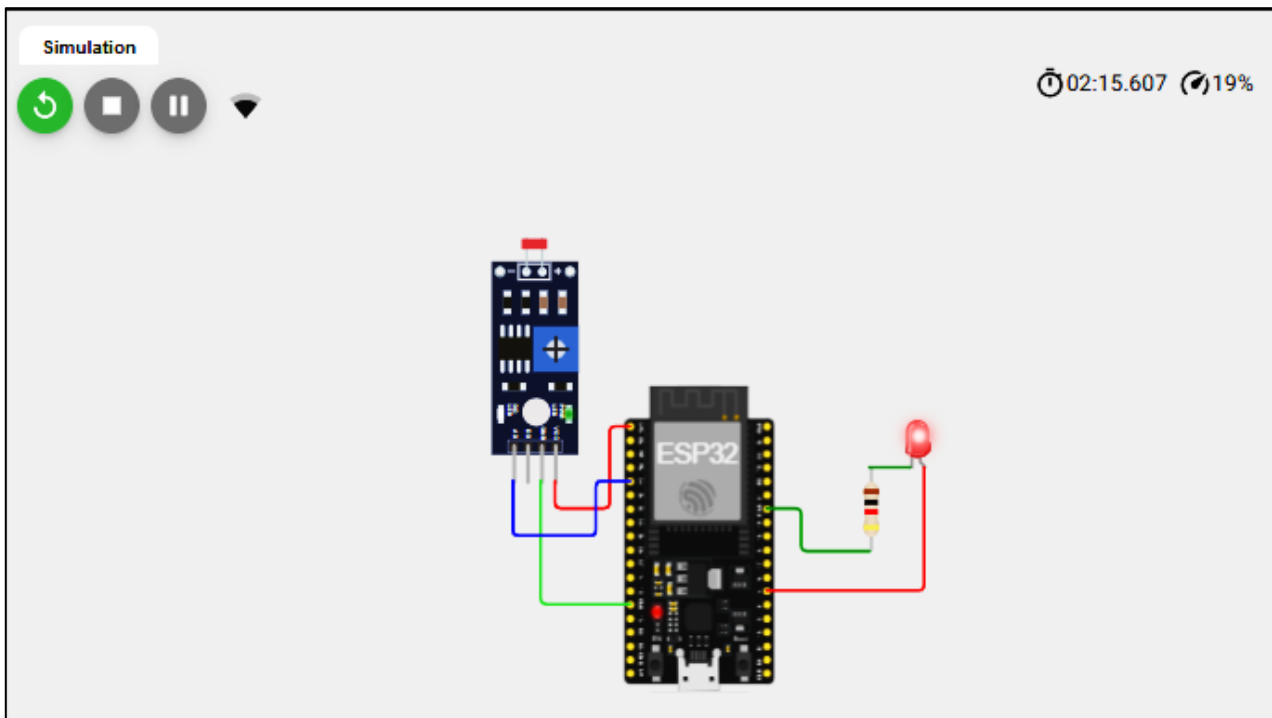


Figure 2 ESP32 circuit running in Wokwi.

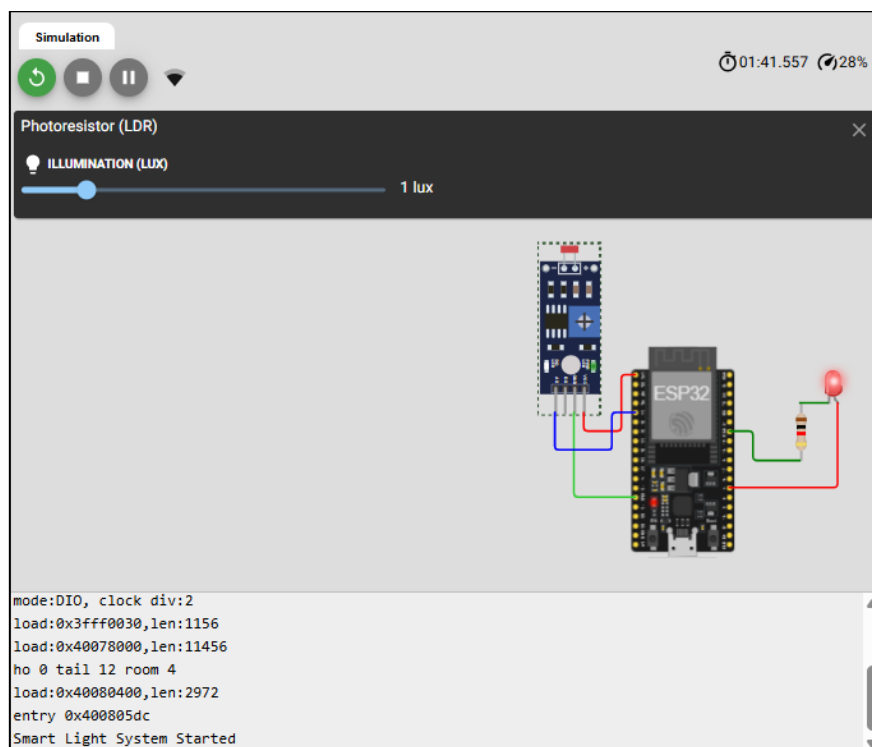


Figure 3 LED ON when environment is dark.

BLYNK WIDGETS DESCRIPTION

1. Manual Light (V0)

This switch allows the user to manually turn the LED ON or OFF from the Blynk dashboard when Auto Mode is disabled.

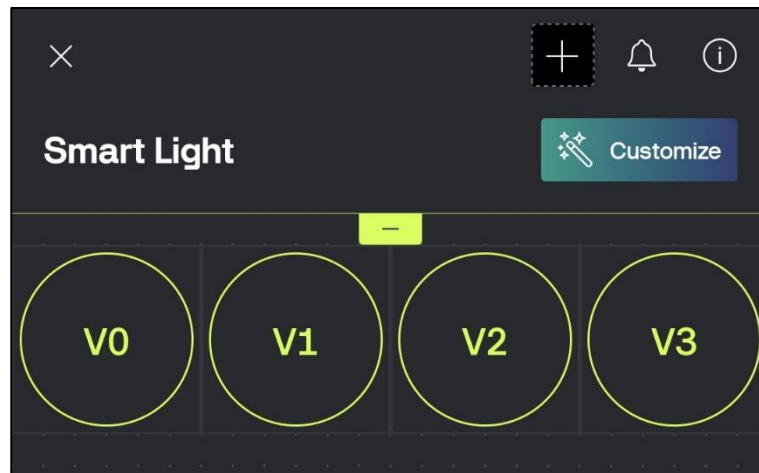


Figure 4 Blynk Dashboard

2. Auto Mode (V1)

This switch enables or disables automatic control. When enabled, the ESP32 controls the LED based on the LDR sensor reading.

3. Light Level (V2)

This gauge/value display shows the real-time light intensity detected by the LDR sensor.

4. LED Status (V3)

This LED widget indicates the current state of the LED. It glows when the LED is ON and turns OFF when the LED is OFF.

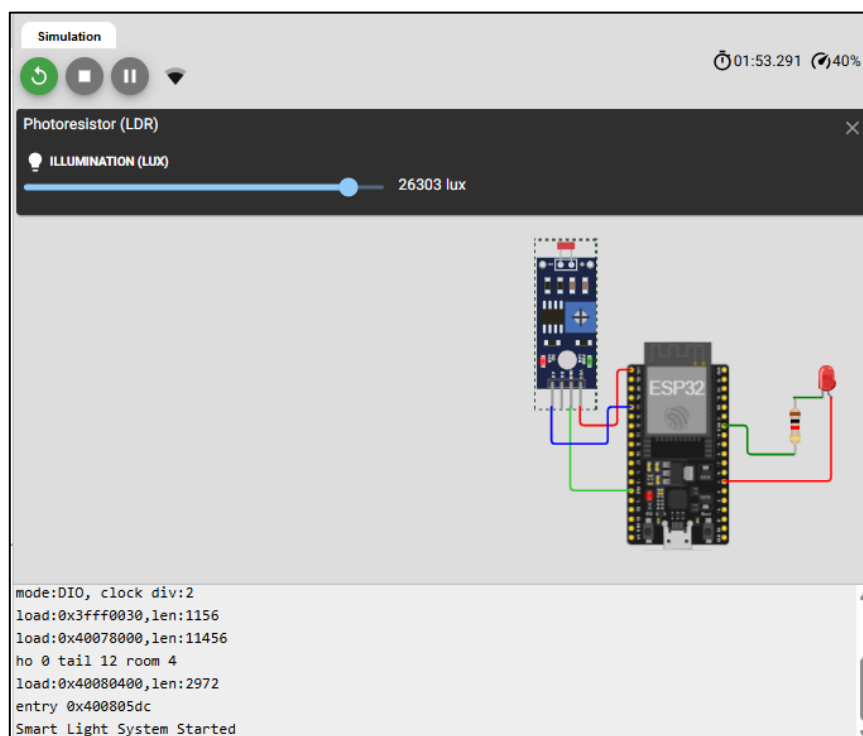


Figure 5 led off when environment is bright.

Conclusion

The Smart IoT Light Control System successfully demonstrates intelligent decision-making, local automation, cloud monitoring, and hybrid control using ESP32 and Blynk. The project improves energy efficiency by automatically controlling lighting based on ambient light conditions while still allowing remote user control.